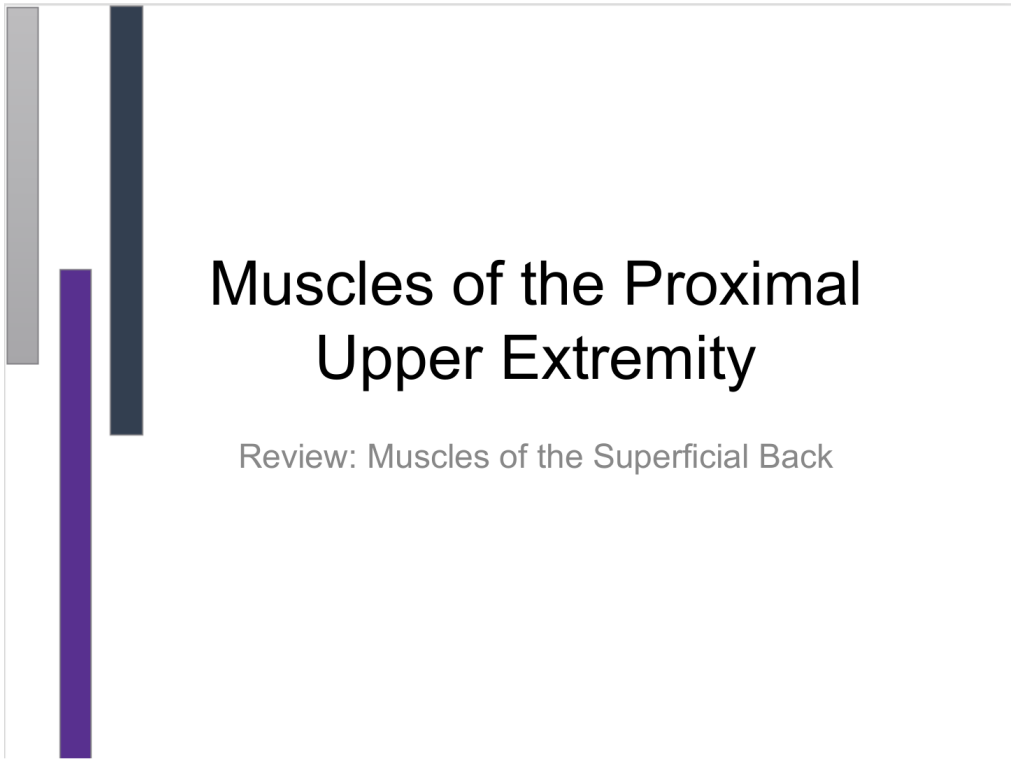
Week #8: [Muscles of the Shoulder]

(Non-hyperlinked = Dr. Ken Faber Module)

- Introduction to Module

Muscles of the Proximal Upper Extremity

- **Review Quiz: Superficial Back Musculature (6 min)**
- **Pectoral Region Musculature (16 min)**
- **Shoulder Region Musculature (24 min)**
- **Approach to Rotator Cuff Disease (21 min)**



Review Quiz: Superficial Back Musculature

Review: Superficial Back Musculature

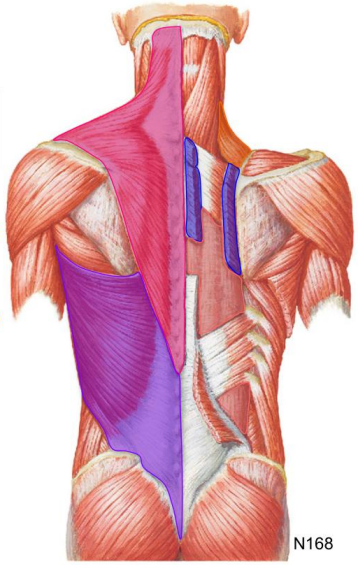
Superficial Musculature

Trapezius

Latissimus Dorsi

Rhomboids

Levator Scapulae



Knowledge Check

Latissimus Dorsi B

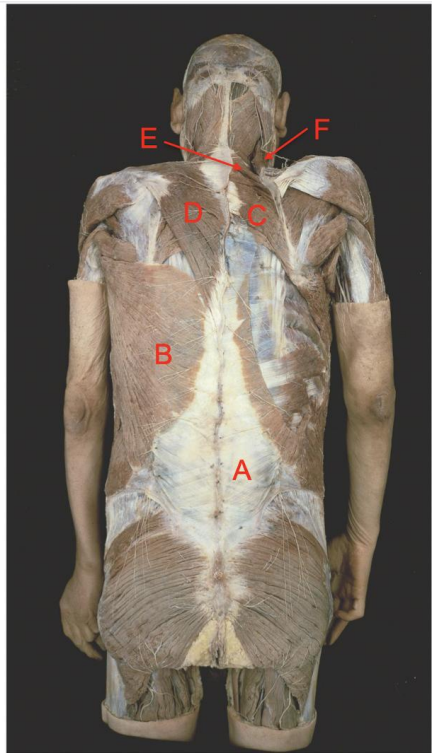
Levator Scapulae F

Rhomboid Major C

Rhomboid Minor E

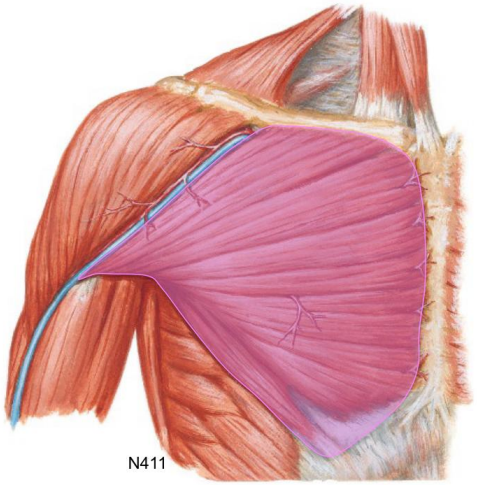
Thoracolumbar Fascia A

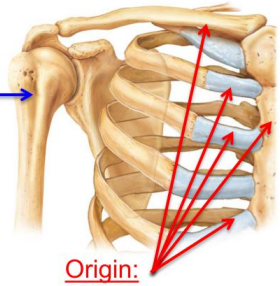
Trapezius D



Pectoral Region Musculature

Pectoralis Major



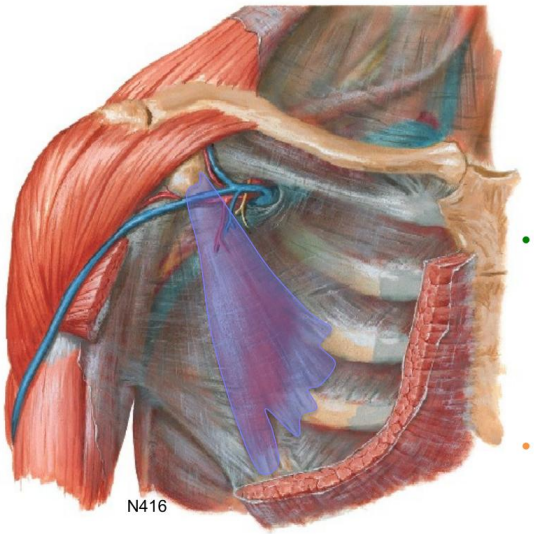


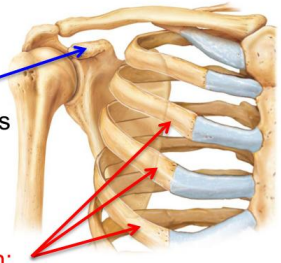
Insertion:
Lateral lip of intertubercular groove

Origin:
Medial Clavicle
Sternum
Upper 6 Ribs and Costal Cartilages

- Function:
  - Adducts, flexes & medially rotates humerus
- Innervation:
  - Medial & Lateral Pectoral Nerves

Pectoralis Minor





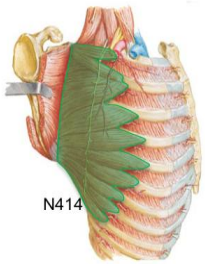
Insertion:
Coracoid Process

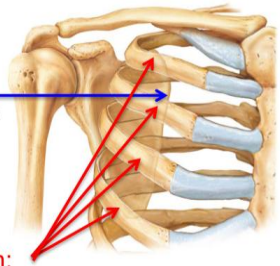
Origin:
Ribs 3-5

- Function:
  - Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
- Innervation:
  - Medial & Lateral Pectoral Nerves

Serratus Anterior







Insertion:
Medial Border of Scapula

Origin:
Ribs 1-8

- Function:
  - Protraction of the scapula
- Innervation:
  - Long Thoracic Nerve (C5,6,7)

Muscle	Origin	Insertion	Innervation	Function
Pectoralis Major	Medial Clavicle Lateral Sternum Upper 6 ribs & costal cartilages	Anterior lateral lip of the intertubercular groove/sulcus	Medial and Lateral Pectoral Nerves	Adducts, flexes, and medially rotates humerus
Pectoralis Minor	Ribs 3-5 (near costal cartilages)	Coracoid Process		Stabilizes scapula by drawing it anteriorly and inferiorly against thoracic wall
Serratus Anterior	Ribs 1-8	Medial Border of Scapula	Long Thoracic Nerve	Protraction of scapula

Clinical Note: If the nerve supply to the serratus anterior is damaged. The serratus anterior is no longer able to hold the scapula against the posterior thoracic wall while pushing with the upper extremity. This results in a **winged scapula presentation.**

Serratus Anterior: Winged Scapula

Long Thoracic Nerve Lesion:

- Will lead to ***winging of the scapula***
  - the serratus anterior is unable to keep the medial border of the scapula against the thoracic wall when pushing with the upper limb

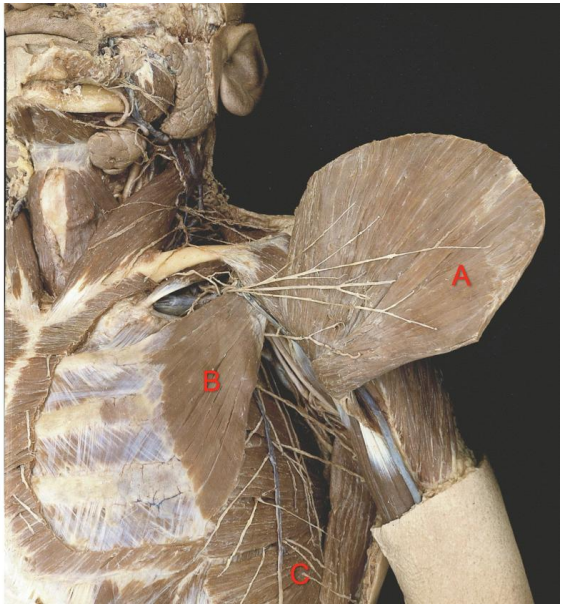


Knowledge Check

Draws Scapula Inferiorly (B)

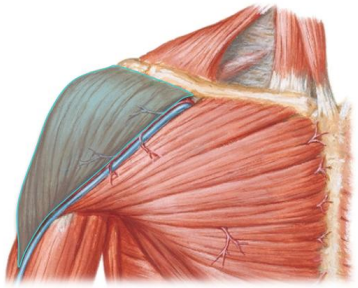
Flexes the Arm (A)

Protracts the Scapula (C)

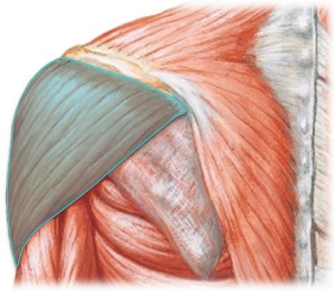


Deltoid

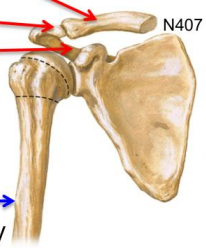
Myotome: C5 & 6



N411



Origin:
Lateral 1/3 of clavicle
Acromion
Spine of scapula

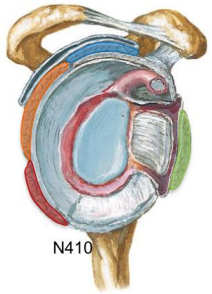


N407

Insertion:
Deltoid tuberosity

- Function:
  - Anterior Head: flexes & medially rotates arm
  - Middle Head: abducts arm
  - Posterior Head: extends & laterally rotates arm
- Innervation:
  - Axillary Nerve (C5, C6)

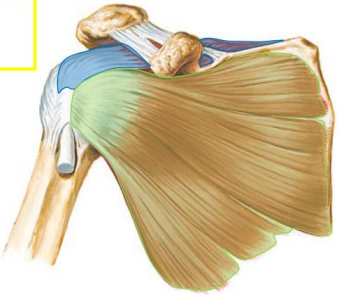
Rotator Cuff



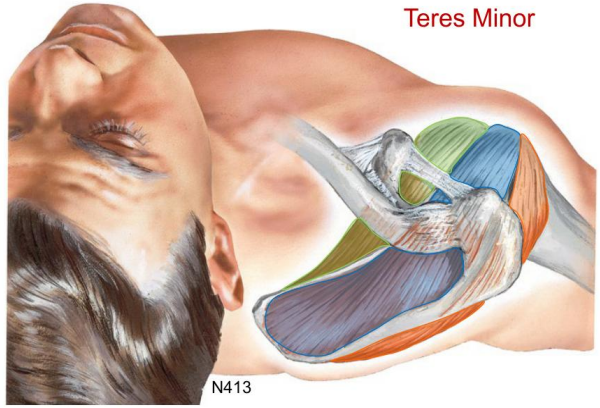
N410

Majority of Rotator Cuff
Myotome: C5 & 6

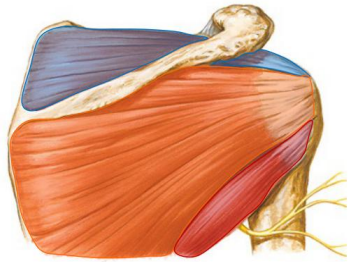
Supraspinatus
Infraspinatus
Subscapularis
Teres Minor



Anterior



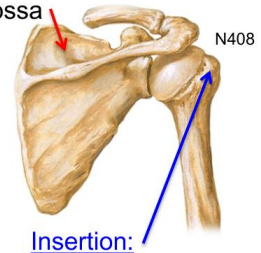
N413



Posterior

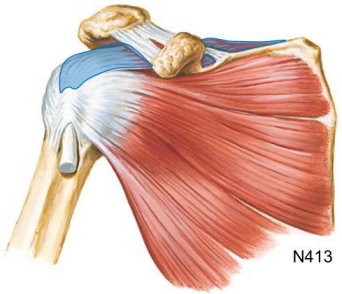
Supraspinatus

Origin:
Supraspinous Fossa



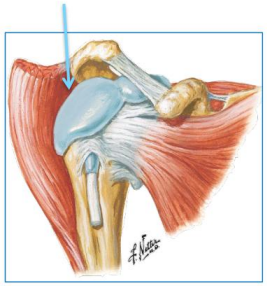
N408

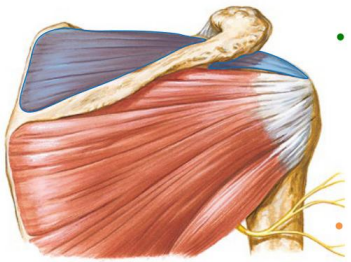
Insertion:
Greater Tubercle



N413

Anterior

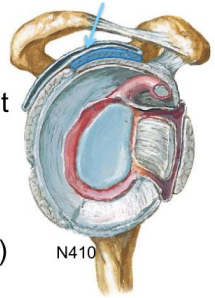




Posterior

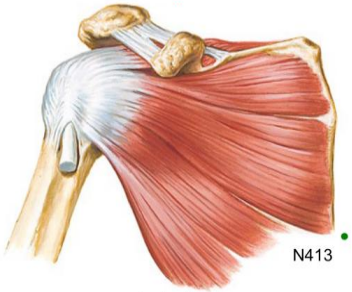
- Function:
  - First 15° of arm abduction
  - Stabilizes glenohumeral joint
- Innervation:
  - Suprascapular Nerve (C5, 6)

Subdeltoid/Subacromial Bursa



N410

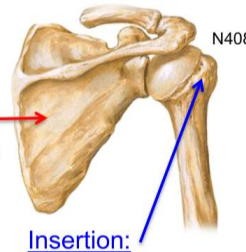
Infraspinatus



N413

Anterior

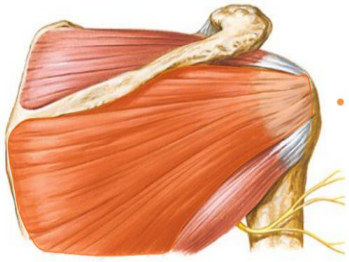
Origin:
Infraspinous Fossa



N408

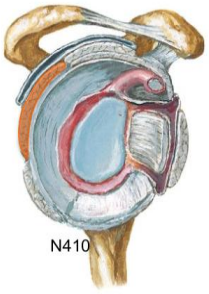
Insertion:
Greater Tubercle

- Function:
  - Lateral rotation of arm
  - Stabilizes glenohumeral joint



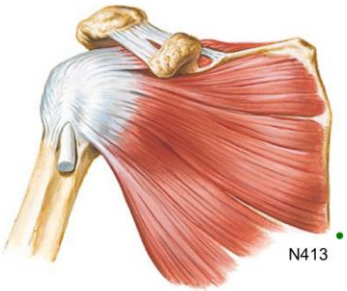
Posterior

- Innervation:
  - Suprascapular Nerve (C5, 6)

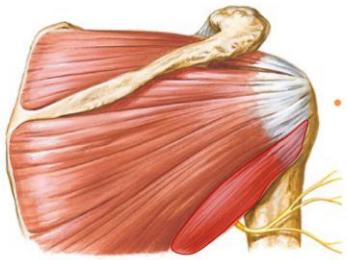


N410

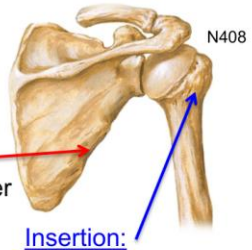
Teres Minor



Anterior



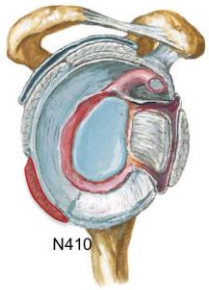
Posterior



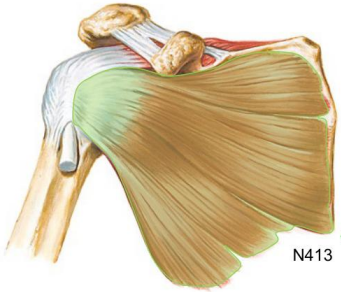
Origin:
Lateral border of scapula

Insertion:
Greater Tubercle

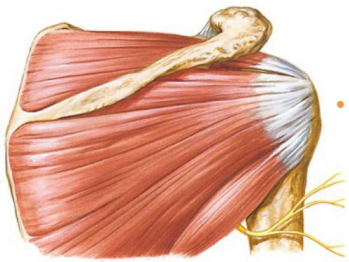
- Function:
  - Lateral rotation of arm
  - Stabilizes glenohumeral joint
- Innervation:
  - Axillary Nerve (C5, 6)



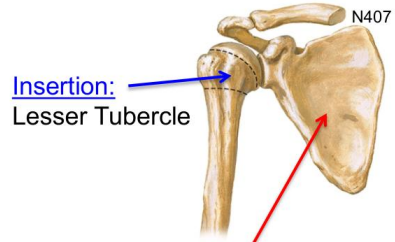
Subscapularis



Anterior



Posterior



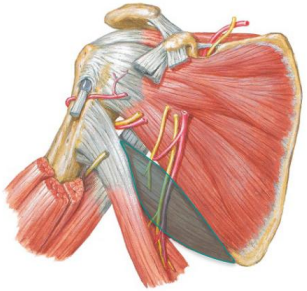
Insertion:
Lesser Tubercle

Origin:
Subscapular Fossa

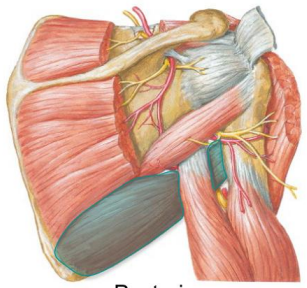
- Function:
  - Medially rotates arm
  - Stabilizes glenohumeral joint
- Innervation:
  - Upper & Lower Subscapular Nerves (C5, 6, 7)



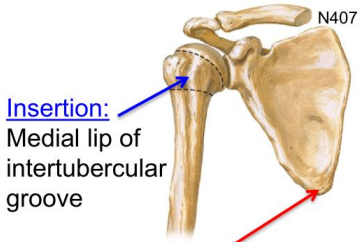
Teres Major



Anterior



Posterior



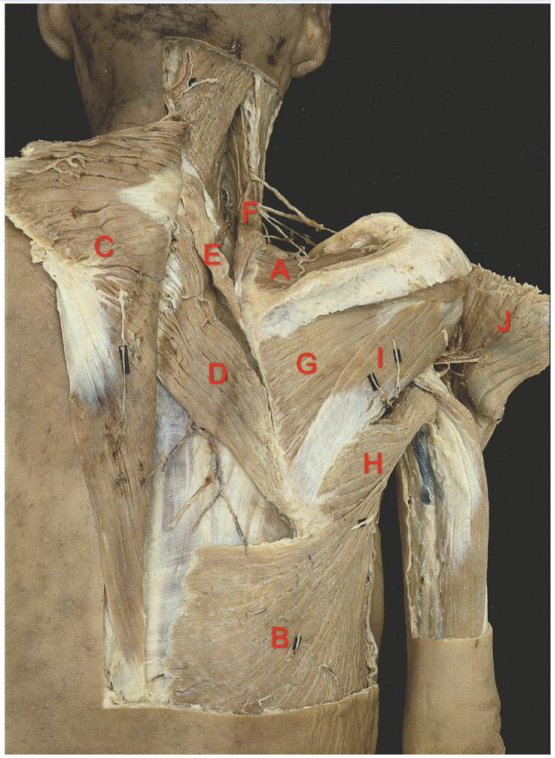
Insertion:
Medial lip of intertubercular groove

Origin:
Inferior Angle

- Function:
  - Adducts and medially rotates arm
- Innervation:
  - Lower Subscapular Nerve (C5, 6)

Knowledge Check

Deltoid (J)
Infraspinatus (G)
Levator Scapula (F)
Latissimus Dorsi (B)
Rhomboid Major (D)
Rhomboid Minor (E)
Supraspinatus (A)
Teres Major (H)
Teres Minor (I)
Trapezius (C)



Supposed to be ADduct

Muscles of the Shoulder Region

Muscle	Origin	Insertion	Innervation	Function
Deltoid	Lateral 1/3 Clavicle, Acromion, Spine of Scapula	Deltoid Tuberosity	Axillary Nerve (C5,6)	Ant: Flexes and medially rotates arm Mid: Abducts humerus Post: Extends and laterally rotates arm
Supraspinatus	Supraspinous Fossa	Greater Tubercle	Suprascapular Nerve (C5,6)	Abduction (first 15 degrees) Stabilizes GH joint
Infraspinatus	Infraspinous Fossa	Greater Tubercle	Suprascapular Nerve (C5,6)	Lateral rotation of arm Stabilizes GH joint
Teres Minor	Lateral border of scapula	Greater Tubercle	Axillary Nerve (C5,6)	Lateral rotation of arm Stabilizes GH joint
Subscapularis	Subscapular Fossa	Lesser Tubercle	Upper and Lower Subscapular Nerves (C5,6,7)	Medial rotation of arm Stabilizes GH joint
Teres Major	Inferior Angle of Scapula	Medial lip of intertubercular groove	Lower Subscapular Nerve	Abducts and medially rotates humerus

ADDuct

Approach to Rotator Cuff Disease

ROTATOR CUFF DISEASE

Kenneth Faber MD, MHPE, FRCSC
Professor of Surgery
University of Western Ontario



ROTH | McFARLANE
HAND & UPPER LIMB CENTRE
ST. JOSEPH'S HEALTH CARE LONDON

Objectives

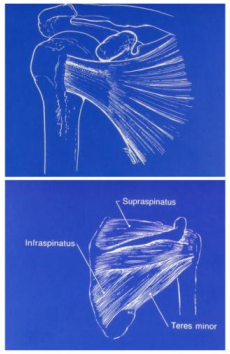
- Review normal rotator cuff anatomy and function
- Define rotator cuff pathophysiology
- Describe available treatments

Rotator Cuff Disease

- Most common shoulder disorder
- Increasing prevalence with age
  - Onset - 40's
  - Considered progressive

Rotator Cuff - Anatomy

- Four muscles
  - Supraspinatus
  - Infraspinatus
  - Teres Minor
  - Subscapularis



Classification

- Stage I
- Stage II
- Stage III
- Rotator Cuff Arthropathy

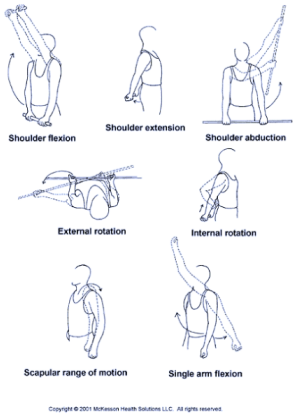
Rotator Cuff Arthropathy

- Massive defect in cuff
- Humerus migrates superiorly
  - Abuts acromion
- Arthritis results



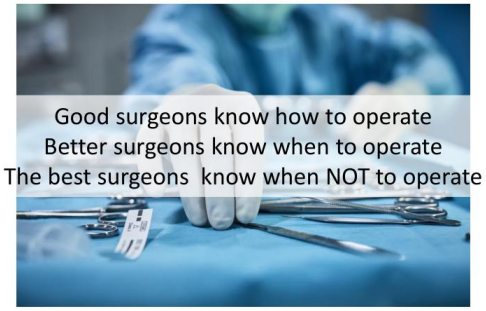
TREATMENT

- Education
- Activity modification
- Pain management
  - NSAIDs
- Physiotherapy
  - Muscle reconditioning
- Injections
- Surgery



? Surgery ?

- When?
  - Pain persists
  - Weakness
  - Following an acute injury



Summary Rotator Cuff Disease

- Very common in middle age
- Usually due to intrinsic tendon changes
- Treatment is non surgical in majority
- Surgery depends on disease severity

